IMAGE: Ashwin Boricha and Jaya Shetty

POWER PROTECT YOUR PC

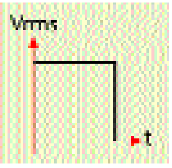
Spare a thought for your electronic equipment and learn to protect it from the vagaries of Indian power conditions

Power failures are something we have all experienced at some time or the other. Whether you live in a metro city where there is relatively 'cleaner' power or in a smaller city with regular power outages, power problems can throw a spanner in the works of our electricity dependent lifestyles. This applies in an even greater degree to large cities—here electricity is the life force behind the equipment that we have grown so dependant upon. Office and home desktop computers, company networks and communication systems such as phones and fax systems—if any of these undergo a downtime during working hours, it spells a substantial loss of money and worse still, data. Besides, with the advent of the monsoons, the reliability of power delivered to our homes and offices reaches higher levels of unpredictability.

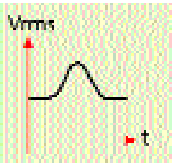
Without power...

Just as in any other combat situation, you need to know what you are up against before choosing your line of defence. There is a general misconception that a power failure comprises of just one situation—a blackout. This is but one of several problems when it comes to power failure. The following are the different types of power conditions that can occur:

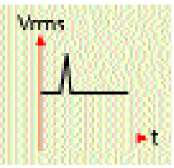
■ **Blackout:** This is when there is a total power failure. It is akin to a light being switched off—the power from your electrical outputs falls from its normal level to zero.



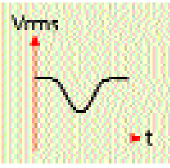
■ **Brownout:** This is a condition where the output voltage drops to a lower level than its normal value (230 volts in India) for an extended period of time. This can affect the operation of electronic equipment and even cause damage to it.



■ **Power surge:** Here, the power level rises above the normal value for a period and then drops to the normal value. This condition can be detrimental to electronic equipment and while most computer hardware can handle slightly higher input voltages over certain periods of time, powerful surges can spell instant death for electronics if they are not protected.



■ **Voltage spike:** This is a condition where there is a brief impulse of a very high voltage after which it returns to the normal level. This condition can be lethal to sensitive electronic equipment including computers. Such conditions occur during lightning storms when a bolt of lightning hits a power line.



■ **Frequency variation:** Here, the frequency of the AC voltage differs from its